

MODEL QUESTION PAPER
JIGS FIXTURES AND GAUGES

*Time: 3 Hour**Max. Marks: 75*

PART A

- I. Answer **all** questions in one word or one sentence. Each question carries **1 mark**.
(9 * 1 = 9 marks)

1	What do you mean by fool proofing?	M1.02	R
2	Name the various operations that can be performed using jig.	M1.02	U
3	List out the purposes of washers in clamping	M2.04	R
4	What is the function of diamond pin locator in machining?	M2.02	U
5	Box jigs are also known as-----	M3.03	R
6	Which type of jig is used for machining in more than one plane	M3.03	R
7	Give an example of a machining process where a grinding fixture is used.	M4.01	R
8	'Go' side of the plug gauge will have the diameter equal to-----of the job	M4.03	R
9	What is wear allowance?	M4.03	U

PART B

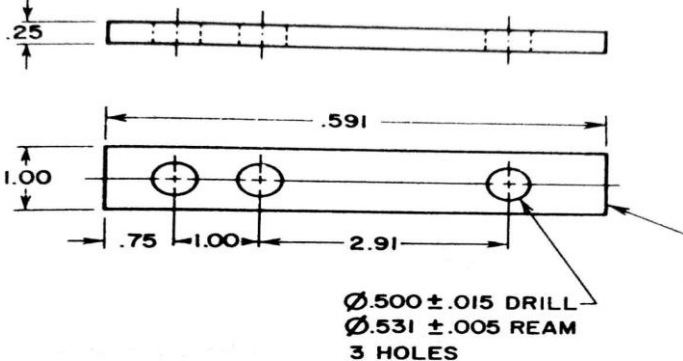
- II. Answer any **eight** questions from the following, each question carries **3 marks**.
(8 * 3=24 marks)

1	Differentiate between Jigs and Fixtures.	M1.02	U
2	Describe about the elements of jigs and fixtures	M1.02	U
3	Draw the neat sketch of wedge clamp.	M2.04	U
4	Describe press fit bushings with neat diagrams	M3.02	U
5	List out the materials for jig bushes.	M3.02	R
6	Write short notes on a plate drill jig	M3.03	U
7	Write short note on modular fixture.	M4.01	U
8	Describe key way broaching fixture	M4.01	U
9	State Taylor's principle of gauging	M4.03	U
10	Describe Receiver gauges.	M4.04	U

PART C

Answer **ALL** questions. Each question carries **7 marks**.

(6*7=42 marks)

III	Describe 3-2-1 principle of location OR	M1.03	U
IV	Compare and contrast the advantages and disadvantages of using jigs and fixtures in manufacturing processes.	M1.02	U
V	List out the general design principles of Jigs & Fixtures OR	M1.04	R
VI	Explain six point location principle in details with figure.	M1.03	A
VII	Briefly explain toggle clamp with neat sketch OR	M2.04	U
VIII	Explain diamond pin locator with a neat diagram.	M2.02	U
IX	What is meant by clamping? Explain the principle involved in the design of clamp. OR	M2.03	U
X	Describe the characteristics and applications of cylindrical locators.	M2.02	U
XI	 Ø.500 ± .015 DRILL Ø.531 ± .005 REAM 3 HOLES Explain Channel drill jig with a neat diagram OR Design and draw a drill jig to drill three holes of diameter 5 for the given figure. Draw minimum two views of jig.	M3.03	U
XII		M3.04	A
XIII	Explain three types of Limit gauges OR	M4.03	U
XIV	Explain with neat sketch milling fixture.	M4.01	U